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Computer Science and Engineering Department

CSE 429: Computer Vision and Pattern Recognition
Project Proposal
Arabic Handwritten Text Recognition using
TrOCR

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Contents

1	Abstract	2
2	Teaser Figure	2
3	Introduction	3
4	Approach	3
4.1	System Input and Output	3
4.2	TrOCR Architecture	3
4.3	Arabic Decoder Modification	3
4.4	Implementation	4
4.5	Challenges	4
5	Experiments and Results	4
5.1	Dataset	4
5.2	Evaluation Metrics	4
5.3	Quantitative Results	4
5.4	Training and Evaluation Curves	5
6	Qualitative Results	5
7	Conclusion and Future Work	6
8	References	6

1 Abstract

Arabic handwritten text recognition is a challenging OCR problem because Arabic script is cursive, characters change shape depending on their position in a word, and handwriting styles vary significantly between writers.

This project fine-tunes **TrOCR**, a Transformer-based encoder-decoder OCR model, on the **KHATT v1.0** Arabic handwritten text-line dataset.

Our system takes an image of a handwritten Arabic text line as input and outputs the corresponding digital Arabic text. The fine-tuned TrOCR model achieved a **CER of 8.10%** and a **WER of 12.50%**, showing a strong improvement compared with the cross-attention Arabic handwritten recognition baseline.

Task	Arabic handwritten text-line recognition
Dataset	KHATT v1.0
Model	Fine-tuned TrOCR
Main Metric	Character Error Rate (CER)

2 Teaser Figure

A handwritten Arabic text-line image is passed through a fine-tuned TrOCR encoder-decoder model, which generates the recognized Arabic text. The system is trained and evaluated on KHATT v1.0 and compared against a cross-attention Arabic handwritten recognition baseline.

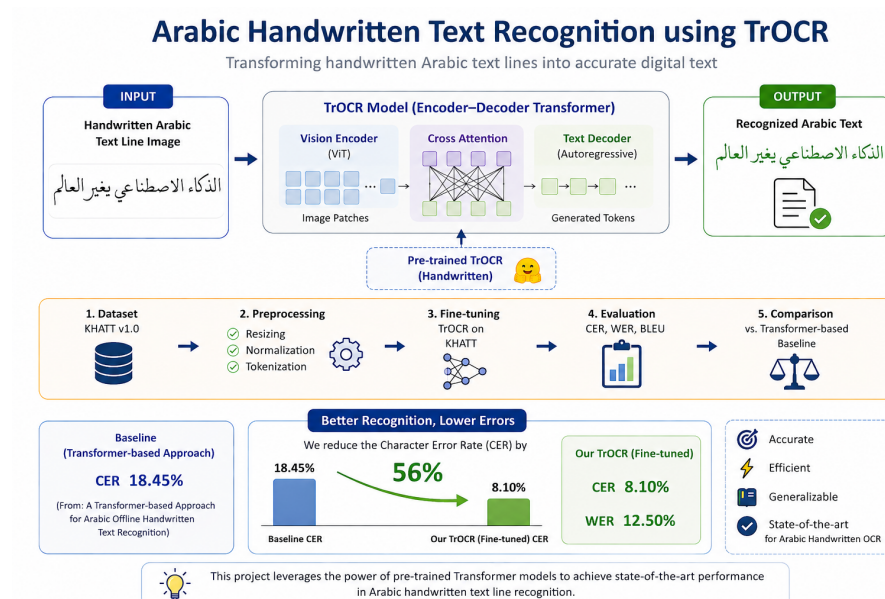


Figure 1: Overview of the proposed Arabic handwritten OCR pipeline.

3 Introduction

Optical Character Recognition (OCR) converts images of text into machine-readable text. While OCR systems have achieved strong performance for printed Latin text, Arabic handwritten OCR remains difficult because Arabic is cursive and highly variable.

Traditional OCR systems usually combine CNNs for visual feature extraction, RNNs for sequence modeling, and CTC for alignment. These pipelines are effective but complex.

TrOCR simplifies the pipeline by using a Vision Transformer encoder and a Transformer decoder in a fully end-to-end architecture.

In this project, we investigate whether a large pre-trained Transformer OCR model can transfer effectively to Arabic handwritten text recognition.

4 Approach

4.1 System Input and Output

The input to the system is a single handwritten Arabic text-line image. The output is the corresponding digital Arabic text transcription.

4.2 TrOCR Architecture

TrOCR is an encoder-decoder Transformer OCR model. The encoder processes the image and extracts visual features, while the decoder generates the output text token by token.

Pipeline Overview

1. **Image Preprocessing:** Input handwritten images are resized and normalized before training.
2. **Vision Encoder:** The image is divided into patches and processed using a Vision Transformer encoder.
3. **Cross Attention:** The decoder attends to visual features extracted by the encoder while generating text.
4. **Text Decoder:** The decoder generates Arabic text autoregressively, one token at a time.

4.3 Arabic Decoder Modification

Since the original TrOCR decoder is mainly designed for English language modeling, we modified the decoder to better support Arabic text generation by initializing it with `bert-base-arabic`.

This modification helps the model better understand Arabic vocabulary, linguistic structure, and contextual relationships during autoregressive text generation. The goal of this change is to improve recognition quality for Arabic handwritten text and reduce character-level prediction errors.

4.4 Implementation

We used Python, PyTorch, and the Hugging Face Transformers library. Existing TrOCR model weights were used as initialization, while preprocessing, fine-tuning, evaluation, and metric computation were implemented for this project.

4.5 Challenges

- **Arabic writing direction:** Arabic is written from right to left.
- **Handwriting variability:** KHATT contains many different writing styles.
- **Character ambiguity:** Many Arabic letters differ only by dots.
- **Evaluation fairness:** CER is used as the primary evaluation metric.

5 Experiments and Results

5.1 Dataset

We use **KHATT v1.0**, an Arabic offline handwritten text-line dataset containing samples from many writers.

5.2 Evaluation Metrics

Table 1: Evaluation metrics used in this project.

Metric	Description	Direction
CER	Character-level edit distance	Lower is better
WER	Word-level edit distance	Lower is better
BLEU	Sequence similarity metric	Higher is better

5.3 Quantitative Results

Table 2: Comparison of our model against the baseline.

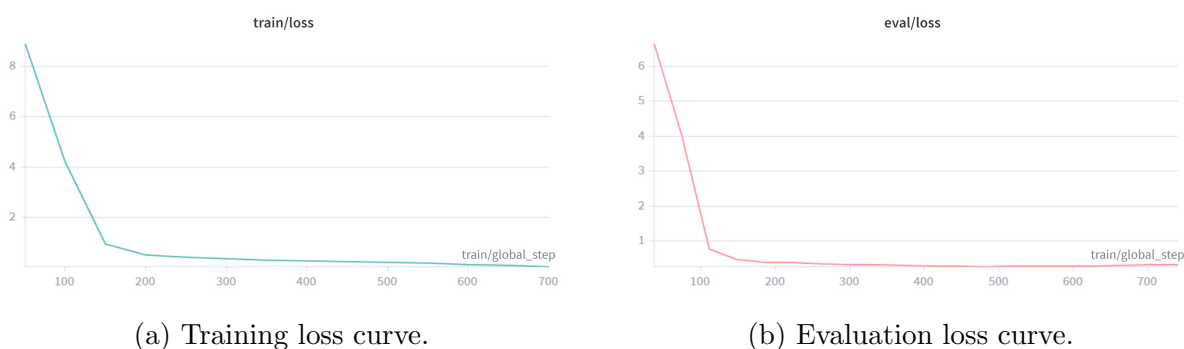
Model	Dataset	CER ↓	WER ↓
Cross-Attention Baseline	Arabic HTR KHATT	18.45%	—
Fine-tuned TrOCR + Arabic Decoder	KHATT v1.0	8.10%	12.50%

The fine-tuned TrOCR model reduced CER from 18.45% to 8.10%, corresponding to approximately a 56% relative reduction in character error rate.

Key numerical results at a glance:

Metric	Value	Source
Baseline CER	18.45%	Cross-attention Arabic HTR baseline
Our CER	8.10%	Fine-tuned TrOCR
Our WER	12.50%	Fine-tuned TrOCR

5.4 Training and Evaluation Curves



(a) Training loss curve.

(b) Evaluation loss curve.

Figure 2: Training and evaluation loss curves during fine-tuning.

6 Qualitative Results

The following examples demonstrate predictions generated by the fine-tuned TrOCR model on handwritten Arabic text-line images.

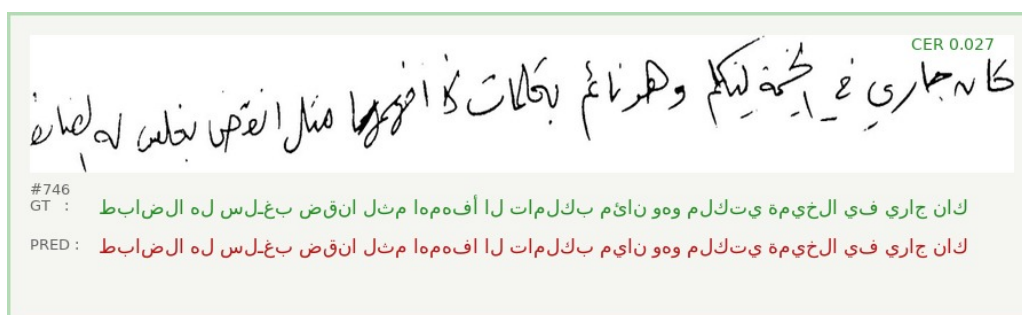


Figure 3: Example predictions produced by the fine-tuned TrOCR model.

Common OCR failure cases include: confusing visually similar Arabic letters, missing dots, connected words, and noisy handwriting.

7 Conclusion and Future Work

This project demonstrates that fine-tuning TrOCR is highly effective for Arabic handwritten text-line recognition. The model achieved a CER of 8.10% and significantly outperformed the selected cross-attention Arabic handwritten recognition baseline.

Future work includes:

- Testing larger TrOCR variants.
- Exploring stronger data augmentation strategies.
- Evaluating on more Arabic handwriting datasets.
- Improving Arabic-specific language modeling components.

8 References

1. Minghao Li et al., *TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models*, 2022.
2. Saleh Momeni and Bagher BabaAli, *A Transformer-based Approach for Arabic Offline Handwritten Text Recognition*, 2023.
3. KHATT v1.0 Dataset.
4. Hugging Face Transformers Library.
5. PyTorch Deep Learning Framework.